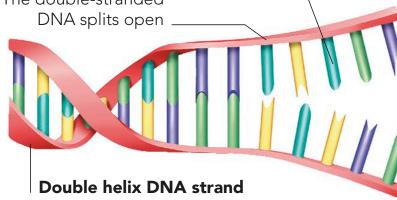
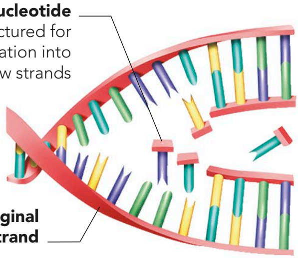
DNA REPLICATION

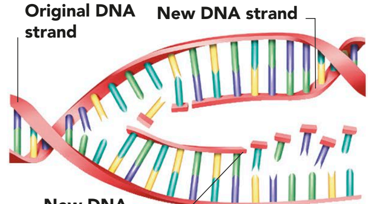
Apart from carrying genetic information in chemically coded form, as its sequences of base pairs, DNA has another key feature. It can make exact copies of itself, a process known as replication, by separating the two backbone strands and the bases attached to them, at the bonds between the base pairs. Then each strand acts as a template to build a complementary partner strand. DNA replication takes place before cell division (see right).



1 SEPARATION
The two strands of the double helix separate at the base pair links. Each base is exposed, ready to latch onto its partner in the newly constructed strand.



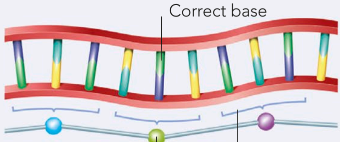
2 BASES JOIN
Free nucleotides, each one a base combined with a portion of DNA backbone, join to the two sets of exposed bases. This can only happen in the correct order, since A always pairs with T, and C with G.



3 TWO STRANDS FORM
More nucleotides join, linked by a new backbone. Each strand now has a new "mirror-image" partner, giving two double helices, which are identical to each other and to the original.

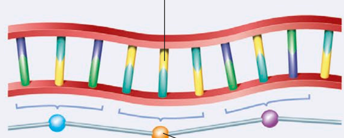
MUTATIONS

DNA replication usually works well. However, factors such as radiation or certain chemicals may cause a fault, where one or more base pairs do not copy exactly. This change is a mutation. The new base sequence may produce a different protein, which could cause a problem in the body.



NORMAL GENE

Each set of three base pairs (a triplet codon) specifies which amino acid should be added to the series of amino acids that make the normal protein for that gene.



MUTATED GENE

In a point mutation, one base pair has become altered and substituted. A different amino acid may be specified, which will disrupt the protein's eventual shape and function.